

The Animal Inside the Machine

Science · 35 min · Felden Vareth · 1 November 2026

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Cyborg insects for rescue, animals used as sensors, and prostheses that restore touch: how biology, technology, identity and agency increasingly intertwine.

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Original article: https://www.eidoslibro.com/blog/articles/10_the_animal_inside_the_machine.html

In June 2026, a group of researchers fitted a cockroach with a tiny diving suit. The covering protected the openings it needed to breathe from the water, while a chemical generator supplied oxygen. Inside its abdomen were an electronic controller and a battery that allowed some of its movements to be directed.

The animal could walk underwater for hours, return to dry land and squeeze through a two-centimetre gap. Its legs, muscles and reflexes came ready-made. Technology extended the abilities of a body that already knew how to move.

For decades, we have built machines inspired by living organisms. We have copied wings, joints, flexible skins and ways of navigating the world. Biomimetics studies these solutions and adapts them to engineering problems. Biohybrid systems take another approach: they incorporate the organism itself.

> *Why build an entire robot when evolution has already built the body we need?*

The idea makes it possible to create small devices with abilities that are difficult to reproduce artificially. It also means intervening in living animals and working with their autonomy. Following these experiments eventually brings us to human prostheses, where a machine can help produce movement, restore a sense of touch and enter into a person's experience of their own body.

Building with Evolution

A Body That Already Knows How to Walk

Building a small robot capable of moving over uneven ground is harder than it looks. Engineers must design actuators, joints, force transmission, sensors, balance control, a power supply and algorithms that determine where each foot should land. Shrinking the robot brings further problems. Batteries take up a substantial share of its volume, there is less room for actuators, every stone becomes an obstacle, and a crack that means nothing to us can become a wall.

A cockroach arrives in the laboratory with much of that work already done.

It has six coordinated legs, an exoskeleton, muscles, mechanical and chemical sensors, reflexes, the ability to travel over complex surfaces and a body shaped to pass through narrow spaces. Much of the coordination of each leg is handled by its own nervous system. Technology adds another layer. Electrodes placed on sensory structures can deliver stimuli that change the direction of movement. An operator sends a signal that influences the animal, and the organism itself handles much of the execution.

The device introduces an instruction into a system that is already moving. The response depends on the species, the condition of the animal and the terrain; steering the insect requires observing that response and adjusting the stimulation. Hence terms such as biohybrid robots, cyborg insects and biorobots. The terminology varies because it describes a new boundary: a system made up of both a living organism and technological components.

An organism grows by taking in nutrients and can repair some of the damage it suffers. The insect obtains the energy for movement from food; its radio and sensors need their own power supply. Using that biological organisation reduces how much has to be manufactured, although the animal must be kept in conditions that allow it to live and perform its task.

In *The Body of Intelligence*, we looked at how a soft hand can adapt to an object through the properties of its materials. In an insect, the body's contribution includes reflexes and local neural circuits as well as mechanics. The controller draws on a relationship between body and environment that was working long before we connected anything to it.

The Machine Inside the Abdomen

The electronic backpack introduced an obvious drawback. A cockroach might fit beneath a surface but then get stuck because the equipment protruded from its body. The device could also snag on objects or take a knock. Some of the very advantage that had made the animal attractive was lost.

In May 2026, Kazuki Kai and colleagues published a controller that could be implanted in the abdomen along with its battery. The controller measured 10 × 10 × 3.5 mm and weighed 0.5 g, excluding the battery. It was powered by a 9 mAh lithium-polymer battery. Placing the components inside the body better preserved the insect's profile and improved its ability to negotiate several of the obstacles studied.

An X-ray from the study shows the controller and battery within the insect's outline: geometric shapes housed in a body that still uses its own muscles and joints. Electrodes connect that equipment to the structures receiving stimulation; a particular mission might also require cameras or other sensors.

When technology is added to an organism, where it sits matters as much as how much it weighs. A load on the back changes the stability and overall height of the system. Inserting it into the abdomen retains some of the advantages of the animal's anatomy, at the cost of an invasive procedure. The same study examined survival after implantation over a follow-up period of up to 120 days.

In the first group, four of the seven cockroaches died within the first week; the remaining three survived for more than eighty days. The authors linked the high early mortality to possible internal damage during insertion and modified the procedure. In a second group, six of seven animals survived for more than fifty days. These are small groups with different survival periods. The results suggest a possible improvement in the procedure that needs confirmation: the difference in survival at seven days did not reach statistical significance. Long-term comparisons with animals without implants also remain to be made.

A Diving Suit for an Insect

Water posed another problem. Cockroaches breathe through spiracles, openings connected to a network of tracheae that distributes gases throughout the body. To keep them active during a dive, the researchers needed to protect that exchange and supply oxygen.

The NTU and Waseda team designed a flexible, waterproof covering fitted to the insect. A chemical generator used hydrogen peroxide and manganese dioxide as a catalyst. The oxygen released was carried through small tubes to the thoracic spiracles. Together, these elements maintained a breathable atmosphere around the protected openings.

In tests published in June 2026, the animals remained active and moved underwater for up to three hours. The researchers also tested routes that involved water and environments containing carbon dioxide. These were controlled conditions designed to explore obstacles that might arise in flooded spaces or places with little oxygen.

The comparison with a diving suit is quite literal. The system brings part of the environment the animal needs to survive along with it. A spacesuit serves a similar purpose for the human body: it maintains conditions compatible with our physiology while we move outside an environment in which we could survive unaided.

In both cases, the ability to act depends on cooperation between the body and its equipment. The organism retains its physiology, while the device temporarily extends the places it can reach. For a rescue cockroach, crossing a flooded area could make it possible to continue an exploration that would otherwise end there. A small break in the terrain, from our point of view, may determine how far an insect can travel and how much ground remains unexplored.

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From the Laboratory to the Real World

Among the Rubble

The most immediate use for this research appears after an earthquake, an explosion or the collapse of a building. A rescue site is almost the opposite of the setting conventional engineering would choose: unpredictable geometry, dust, darkness, unstable materials, irregular gaps, water, obstacles and difficult communications. A wheeled vehicle can be stopped by a chunk of concrete. A drone needs space for its propellers and creates air currents. A legged robot can overcome some obstacles, but the smaller it needs to be, the harder it becomes to fit adequate motors, batteries and sensors inside.

Cockroaches have spent millions of years moving through narrow, dark and cluttered environments. On 30 March 2025, ten cyborg cockroaches developed by the NTU team travelled to Myanmar with a Singapore Civil Defence Force contingent following the magnitude 7.7 earthquake two days earlier. According to NTU, this was the first use of such insects in a real humanitarian operation.

The deployed insects still belonged to the generation fitted with external backpacks, designed to carry sensors into gaps inaccessible to a person or a larger robot. NTU's published account describes their participation in Myanmar and lessons for future deployments, but provides no assessment from which their effectiveness at finding survivors can be measured. One need identified by the team was the ability to prepare large numbers quickly: manual assembly could take more than an hour per insect.

Building a Swarm

In July 2025, the same group presented an automated system capable of fitting the electronic backpack and electrodes to a Madagascar hissing cockroach in approximately one minute and eight seconds. It uses computer vision to identify the anatomical insertion points and position the components. According to NTU, four insects were equipped in less than eight minutes. The team also developed a backpack that required a lower voltage to deliver stimulation.

Preparing many insects in a short time would make it possible to explore several routes of entry at once. The group's readiness would also depend on animal care, equipment checks and reliable assembly. Any time saving would need to be maintained without increasing injuries or procedures that have to be repeated.

The cost of a cyborg cockroach goes far beyond the animal itself: electronics, sensors, preparation and maintenance all have a price. These prototypes have no comparable commercial price per unit. Automation may make assembly cheaper, but the cost of a mission also depends on personnel, communications and equipment recovery. Evolution has produced the body; engineering bears the cost of intervening in it, connecting it and adding the functions it needs.

The logic increasingly resembles that of a fleet. A swarm can divide the work: some insects carry cameras, others thermal sensors or chemical detectors, while still others can act as communications nodes. Losing one individual does not necessarily mean losing the entire mission. Swarm robotics has long studied how groups of relatively simple agents can cover an area, divide tasks or maintain a distributed network. In these projects, each animal retains its own locomotion while technological coordination attempts to guide the group's work.

Cockroaches Carrying Medicine

In August 2026, researchers at the University of Queensland and UNSW published the so-called *Paraborgs*, based on the giant Australian burrowing cockroach, *Macropanesthia rhinoceros*. Its size and strength allow it to carry a camera or a small injection mechanism.

The proposal combines one insect that provides the operator with video images and another that carries the injector. In laboratory testing, the full sequence of navigation, positioning and injection achieved a success rate of approximately 72%. In short-range injection trials, over distances of up to fifteen centimetres, the rate reached 95%. Those figures come from different

experimental conditions and do not yet allow us to estimate how effective an intervention would be for someone trapped beneath debris.

The researchers tested the injection mechanism on silicone and pig skin. Before it could be used in a real medical intervention, questions of navigation, patient identification, sterility and dosage would still have to be resolved. Getting close to the target is not enough, either: the animal must be correctly oriented and stable for the injector to work.

The researchers themselves envisage specialised swarms. Some insects would locate victims; others would carry medical equipment or environmental sensors. Critical decisions would remain in human hands, while the system would extend the rescuer's physical reach.

Are These Animals Disposable?

In systems with external backpacks, the equipment can be removed after use. NTU states that its modules can be taken off during long rest periods or when the insect is no longer being used, without adverse effects from removal. The Australian Paraborg team also uses detachable harnesses and reports that the insects can live as long as other specimens once the equipment is removed. These are statements made by the research teams about their own procedures.

The fact that a backpack can be removed tells us little about electrode injuries, subsequent recovery or what ultimately happens to the animal. Abdominal implants add surgery and the mortality already described. Evaluating these techniques means looking at how the insect feeds and moves after the intervention, as well as how long it survives.

In a real mission, whether the animals can be retrieved also matters. Equipment that can be removed in a laboratory may not be removable from an insect lost among the rubble. That fate is part of the cost of using living beings as exploration devices.

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When an Animal Becomes Technology

Beyond Cockroaches

The combination of animals and electronics extends to many other species. Flying beetles have been used to study remote control of take-off, landing and direction. In work by Hirotaka Sato and colleagues at Berkeley, for example, small electronic systems and electrodes were fitted to act on muscles and structures involved in flight. A beetle gives a miniature device its wings, muscles and aerodynamic control: a complete biological flight system that electronics can influence.

Experiments have also involved moths. Some projects even made use of metamorphosis, introducing interfaces during the pupal stage so that certain connections would remain available after the adult insect emerged. In the DragonflEye project, Draper and HHMI's Janelia research centre investigated ways of combining miniaturised electronics with optical stimulation of neurons involved in dragonfly flight.

Locusts offer another possibility: making the animal's own sense organs part of the device.

When the Animal Is the Sensor

We have built devices that perform functions comparable to those of our own organs: cameras that capture images, microphones that record sound, actuators that generate movement. Here the animal's sensory system is retained, and an interface is built to read its activity.

> *We can build an artificial sensor, or learn to read electronically a sense that evolution has already created.*

The conceptual difference becomes clear when we compare a camera mounted on a cockroach with the sense of smell of a locust connected to electrodes. In the first case, the animal carries a sensor that we designed. In the second, its chemical receptors, neurons and part of its own processing form the sensor. Researchers add recording and decoding. In experiments at Washington University, locusts distinguished vapours associated with TNT, DNT, RDX, PETN and ammonium nitrate, and their neural responses could be differentiated within a few hundred milliseconds.

For the system to work as an instrument, its stability, differences between individuals and possible confusion between odours must be studied. The interface makes a biological capacity accessible, but that capacity still belongs to a living being.

The distinction between organism and device also emerges if we return to the rescue cockroach and ask what, exactly, is acting. Biologically, there is a cockroach; for the mission, there is a system comprising the animal, sensors, software, communications and operator. If any of those elements fails, the function we were seeking may disappear even though the cockroach remains alive. The organism's identity and the continuity of the functional system follow different criteria. This separation hardly troubles us when we are talking about an insect. Later, when we apply the question to a person with devices connected to their nervous system, the distinction will no longer seem so straightforward.

Land, Air and Water

Caltech has developed bionic jellyfish intended for ocean exploration. Small electronic devices stimulate their swimming, while an added structure on the bell allows them to carry sensors. Their locomotion offers a way to study a low-power platform for transporting instruments, although the payload changes the conditions in which they swim.

There are also cyborg rats. Research published since the beginning of this century has involved implanting microelectrodes in brain regions capable of producing signals that the animal learns to interpret as directional cues and rewards. In 2019, researchers even demonstrated a brain-to-brain interface in which signals extracted from a person were converted into commands sent to a rat equipped with neural stimulation.

Pigeons represent a further step. Different groups have studied how brain stimulation can influence flight direction, how position can be recorded using GPS, and how those records can be used to investigate strategies for correcting a flight path.

The level of development varies greatly between projects. A pigeon guided during an experiment is a long way from a fleet of birds carrying out autonomous missions. Even so, the range of studies shows that the choice of body determines much of what the system can do: flight, access to tunnels and the ability to remain in water depend on different biological capacities.

Making use of an organism also means working with its needs and with differences between individuals. The environment, the required reliability and the cost all influence which bodies might be suitable for a given mission.

The Other Side of Rescue

Technology designed to find someone beneath rubble could also be used to observe a private or inaccessible space. A small insect has characteristics that make it attractive for surveillance: it is hard to detect, produces little noise, can hide in narrow gaps and uses its own muscles to move, reducing the electrical energy required for locomotion. If it also carries a camera, microphone, chemical sensor or radio link, what the technology means depends largely on who sends it and why.

The relationship between biohybrid research and defence has clear precedents. During the 2000s, DARPA launched HI-MEMS, *Hybrid Insect Micro-Electro-Mechanical Systems*, to explore the integration of microelectromechanical systems with insects. Some subsequent work on controlled flight and sensors grew out of that research ecosystem. The study of explosive-detecting locusts received funding from the US Office of Naval Research. A capability presented in terms of detection and safety may also have military relevance.

This is the problem of dual use. Locating people, detecting gases, inspecting tunnels, mapping interiors, carrying small loads and keeping sensors in places that are difficult for humans to access are valuable functions in an earthquake, a mine, an industrial fire or a war. The purpose of the mission determines what information is gathered and who is affected by the intervention.

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The Ethical Problem

Cockroaches arouse little empathy. We readily accept procedures performed on them that would disturb us if the animal were a dog or a primate. That difference in our reaction affects what we are prepared to permit, although it tells us little about what the animal experiences.

We know that insects have nociceptive mechanisms capable of detecting harmful stimuli and producing responses. Determining what kinds of experience might accompany those mechanisms is harder. This uncertainty matters when assessing injuries, the intensity of stimulation and the duration of procedures. It also means looking beyond the animal's ability to perform the task assigned to it.

Regulations distinguish between groups of animals. European Directive 2010/63/EU on animals used for scientific purposes covers live cephalopods alongside the vertebrates within its scope. This is a significant distinction among invertebrates: an octopus and a cockroach are treated differently under that framework. Legal categories define obligations, while research continues to examine the capacities of individual species.

The biological closeness that makes many vertebrates useful for studying our own bodies also makes the consequences of using them harder to ignore. That resemblance can heighten our concern, although an animal's capacity to suffer does not depend on how much of ourselves we recognise in it.

The organism's technological appeal and the ethical problem arise from the same source. We choose an animal because it moves, senses, obtains energy and responds to its surroundings with an efficiency that is difficult to reproduce. The most sophisticated component of the system is still a living being. Its capacities evolved in the service of its own survival; we recruit them for a mission that has nothing to do with it.

> *The fact that the animal is alive is what makes it useful, and also what obliges us to consider what we are doing to it.*

Would we accept using a hundred implanted insects to find a person buried under rubble? Would we accept the same intervention to surveil a room? Change the purpose, and our judgement of a technically identical procedure changes with it. If the animals were rats or pigeons, we would also have to consider different evidence about pain, learning and behaviour.

Using anaesthesia during an intervention, reducing stimulation and removing equipment whenever possible are important measures. But the assessment begins earlier, when we decide whether using the animal is justified and whether an alternative could perform the same function with less harm. Ethics accompanies engineering from the moment we choose the organism that will become part of the device.

Imagine that the technology improves to the point where interventions cause no detectable harm. The animal continues to eat, move and live as long as others of its species. Those results would count in favour of the procedure, but their significance would depend on what we had

measured and how sensitive our tests were. Failing to detect harm and being able to say that there is no suffering are different things. The first statement describes the outcome of an observation; the second concerns what the animal experiences. Survival and the ability to perform a task are not, on their own, enough to bridge that gap.

Uncertainty does not prove that the animal is suffering, either. It requires us to ask what evidence we would consider sufficient to authorise an intervention, and how much risk of being wrong we would accept. Would it be enough that nobody had demonstrated suffering, or should we require positive grounds for considering it unlikely that the procedure causes it? The answer affects who must justify the use of the animal and which alternatives we would be willing to try first. That requirement should reflect the available evidence, the potential harm and the purpose of the intervention.

We can take the argument a step further and suppose, solely to examine the moral problem, that we could genuinely guarantee the absence of suffering and other harms. Would using the animal then be justified? For some ethical positions, eliminating those harms would resolve the main objection. Others would attach value to the organism being able to live and behave without being subordinated to someone else's objectives. Is ensuring its welfare enough to turn it into an instrument? At this point, the disagreement concerns the obligations we believe we have towards other living beings, beyond what we can measure.

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From Animals to Ourselves

A Hand That Gives Back the Sense of Touch

In the systems described so far, a person sets the task and uses technology to intervene in another organism. Human neuroprostheses connect devices to the intentions of their own users. A person with paralysis can learn to control a cursor or a robotic arm using signals recorded from their brain.

The cochlear implant provides a familiar example of such a connection to the nervous system. A microphone picks up sounds, a processor converts them into signals, and electrodes stimulate the auditory nerve. With training, many people can interpret these signals as sounds and speech. The hearing they gain has its own characteristics, and outcomes vary between users.

Brain-computer interfaces use patterns of activity associated, for example, with attempting to move a limb. Electrodes record some of that activity, and software learns to associate it with actions. The intention to reach for an object can then be translated into a command for an artificial hand. Performance depends on training, signal quality and the stability of the interface.

Neuralink is one of the better-known projects in this field, using implanted threads to record neural activity. Other approaches, such as Synchron's, collect signals through a device placed in a blood vessel near motor regions of the brain. Academic teams have studied a variety of solutions for years. The differences between them affect the surgery required, the information they can capture and the conditions in which they can be used.

Movement is only part of our relationship with a hand. When we button a shirt, touch and information about the position of our fingers help us adjust the movement. That feedback lets us handle a button without watching every detail. A prosthesis without sensory feedback forces its user to rely much more heavily on sight.

The University of Pittsburgh's neuroprosthetics programme has combined recordings from the motor cortex with stimulation of the somatosensory cortex. Sensors in a robotic hand detect

contact, and the system converts it into brain stimulation. One participant was able to identify which robotic finger had been touched while blindfolded.

That feedback creates a loop between action and perception. The person tries to move the hand, the device responds, and contact returns to them as sensory information. Learning involves both controlling the movement and interpreting the feedback. The artificial hand can begin to take a stable place in everyday tasks.

Some systems also aim to restore movement to a person's own body. An injury may interrupt communication between the brain and the circuits that move a hand, even though the intention to move it remains. These interfaces seek to establish an alternative pathway by recording brain signals and stimulating the nervous system.

A study published in *Nature Medicine* in July 2026 described a neuroprosthesis linking a brain interface to stimulation of the spinal cord and cerebral cortex. In one participant with complete tetraplegia, it enabled control of hand movements and improved their ability to perform everyday tasks. Persistent improvements in certain motor and sensory functions were also observed, even when the system was switched off.

This was a result in a single person, and more research is needed to establish how far it can be generalised. The work combines immediate assistance with mechanisms intended to encourage neuroplasticity. Technology participates in movement while researchers investigate whether it can help reorganise nervous-system functions over the course of training.

Repair or Enhancement

For someone who has lost mobility, being able to communicate or use a hand can change their everyday independence. The medical value of these interfaces becomes clear in specific tasks: choosing a word on a screen, holding an object or bringing food to the mouth. Research must also establish how to sustain these abilities over months and years, outside the controlled environment of a laboratory.

The same techniques make it possible to investigate control of additional devices. Research into supernumerary effectors explores how extra robotic limbs can be coordinated while the person's natural limbs remain in use. A study published in *Nature Communications* in 2026 examined a non-invasive interface in ten participants without motor impairments. Using electroencephalography, it recorded responses to tactile stimuli on which the user was focusing their attention. The system made it possible to test control of additional robotic arms while participants continued to use their natural arms.

Learning to use an extra limb involves working out how to issue commands and how much the task interferes with ordinary movement. Being able to operate a device in an experiment does not mean it will feel like part of the body. Someone can use a tool skilfully and still experience it as something external. To study its incorporation into bodily experience, we need to ask what the person feels and how they relate to it.

Additional sensory channels could also be introduced. An infrared camera or a chemical detector provides information that our senses register differently, or cannot detect at all. The challenge is deciding how to present those signals to the user, what they can learn to distinguish and how much effort interpretation requires. Having a better sensor is only one part of the process.

Bodily enhancement would also raise questions about access, maintenance and dependence on manufacturers. A function used every day would need support, repairs and guarantees of continuity. For the user, a breakdown or faulty update could directly affect how they move or perceive their surroundings. The experience of having a technologically enhanced body would include that material dependence.

Someone could regain the independence to eat, work or leave home while also depending on a company to retain those abilities. If the manufacturer stopped supporting the device, what guarantees should its user have? When a technology plays a daily role in movement or perception, decisions about its repair and continued operation affect the control a person

retains over their own body. Gaining the ability to act and becoming less dependent on others do not always go hand in hand.

When a Prosthesis Becomes Part of the Body

Imagine someone who has used a robotic hand capable of providing sensory feedback for years. There is a glass on the table. They try to pick it up, and the hand moves towards it; when the fingers close around the glass, they receive a signal of contact. At first, they have to watch and make corrections. With training, they may learn to adjust their grip without monitoring every movement.

The device responds to an intention and returns information that allows the action to continue. Over the years, that relationship may become so familiar that the person no longer thinks about the equipment each time they use it.

> *When they say “my hand”, the phrase may cease to be a metaphor.*

This scene brings together possibilities under investigation and assumes a period of prolonged use that has yet to be studied. Bodily incorporation varies between people and devices. Effective control, a sense of ownership and the experience of being the one performing an action are not exactly the same thing, either. A prosthesis may be useful while continuing to feel strange. Under other conditions, predictable movement and sensory feedback may help it become integrated into the person's representation of their own body.

The material still matters to how the device works, but on its own explains little about that experience. What counts for the person using it is the contact they feel, the confidence with which they act and the continuity between wanting to do something and being able to do it.

An anatomical boundary is easy to draw. Bodily experience is harder to pin down. A walking stick used every day can become part of how we judge distances; an artificial hand that restores touch opens up a much deeper possibility. Its user might recognise movements made through artificial fingers and electronic sensors as their own. Which criterion should then determine where their body ends: the biological origin of its parts, the connection to the nervous system, or the experience of the person using them? The answer may change depending on whether we are discussing anatomy, perception or personal identity.

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When the Machine Makes Decisions Too

Control of a rescue insect can be distributed: a person sets the mission, a program selects part of the route, and the animal handles locomotion. Describing each contribution helps pinpoint errors and identify what needs improving. When a person uses a neuroprosthesis, another factor enters the picture: the experience of someone who recognises, or no longer recognises, a movement as their own.

A prosthesis that interprets neural signals works with incomplete information. It records part of the brain's activity and converts it into commands. To make an action easier, the controller can work out details that the user barely specifies: adjusting a trajectory, regulating force or correcting the stability of a grip.

Imagine that someone wants to reach for a glass and the system recognises the object. The user supplies the intention; the program calculates part of the movement and adjusts the fingers when contact occurs. That assistance can reduce the effort required to use the prosthesis. The

action still serves what the person wanted to do, although some of the adjustments take place outside their nervous tissue.

In *Spider-Man 2*, Doctor Octopus's mechanical arms contain an artificial intelligence that eventually begins to influence him. The fiction takes to an extreme a question that also arises here: how much autonomy do we give a device that interprets and carries out our intentions?

When that assistance fails, the user needs to understand what happened and have safe ways to regain control. Trust in the device also depends on being able to recognise its limits.

In a future visual neuroprosthesis, AI could process images from a camera before converting them into stimulation of the nervous system. Highlighting obstacles, estimating depth or reducing noise would change the information reaching the user. If the user came to experience that information as vision, artificial processing would become part of their everyday perception. Studying AI's role in that perception does not require us to settle whether the program itself has conscious experience.

If AI selects what to emphasise, reconstruct or discard, it also shapes the information available for making decisions. Someone might choose carefully among the options they perceive, unaware that the system has concealed a relevant fact or presented an estimate as though it were an observation. The decision would still be theirs, but it might have been different had that information been available. The effect would depend on how much the device filters and on the opportunities to check what it presents.

Agency is the capacity to act and influence what happens; the sense of agency is the experience of being the one performing an action, and it may persist even when we do not know how much the device is contributing. Filtering information affects the conditions under which we choose and connects with the debate about free will, although it does not settle it. In the case of a prosthesis, the problem is concrete: the user can deliberate and choose, but only among alternatives that another system has selected in advance. Assistance that lets them detect previously invisible obstacles could expand their ability to act; assistance that leaves out relevant information could restrict it without their knowing. Understanding what the device changes, and being able to review or correct that filtering, would become part of having control over one's own life.

Our own bodies make countless adjustments outside conscious deliberation. We walk without calculating every muscle contraction and correct small losses of balance before making a conscious decision. We can experience an action as our own while knowing little about many of the operations that make it possible.

That helps explain how artificial assistance might become part of movement. If a prosthesis spends years anticipating, correcting and completing actions in a predictable way, its user may stop experiencing each intervention as something separate. Familiarity grows through use, and it can also be broken when the device responds unexpectedly.

The expression “shared agency” describes this division of participation. Its extent depends on which functions the device takes on. Sharing control does not, by itself, demonstrate the existence of shared consciousness.

Completing a movement, preventing it and selecting the goal of an action give the device different degrees of involvement. A prosthesis programmed to block a dangerous gesture may stop its user from doing what they want without changing what they desire. If the system independently selects which object to pick up, it also intervenes in the immediate goal. Imagine that the user attempts a movement and the prosthesis blocks it to prevent an injury. The machine may be right about the risk. Should it also have the final say? How far would we be willing to surrender control over our own bodies to a system we consider more competent than ourselves?

Responsibility raises another problem. Consider an artificial hand that breaks an object because the algorithm miscalculates the force needed to hold it. The person decided to pick it up, the manufacturer designed the control rules, and the program determined part of the movement. The user may have experienced the action as their own while being unable to prevent the error.

Legal responsibility would depend on the circumstances and the applicable rules, but a philosophical question remains: how much control must we exercise over an action to consider ourselves responsible for it? The body's own automatic processes already complicate that attribution. An artificial device adds design decisions made by other people, which the user may neither know about nor be able to change.

What Remains When the Body Changes

The Great Transfer imagined in *Eidos* takes these questions much further. It envisages a person continuing to exist in a different substrate. Today's interfaces let us study partial changes: an assisted motor function, an artificial sensory pathway or a new way of acting on the environment. To consider a complete transfer, we would have to extend the question to the processes that sustain our memories and experience.

One hypothesis I explore in Eidos places personal continuity in the organisation of processes and their relationships over time. Gradual replacement allows us to imagine a change of substrate while the system continues to function. Even so, preserving behaviour and memories would not be enough to demonstrate that the experience still belongs to the person who began the process.

Imagine that it were possible to replace a neuron with an artificial device capable of reproducing all the relevant functions and relationships it maintained with the rest of the brain. We then replace another, and another. Suppose that after each intervention the person retains their memories, personality and ability to recognise themselves. As part of the thought experiment, let us also preserve the other conditions necessary for the brain to function. If we continue until every neuron has been replaced, at what point could we say that they are no longer the same person? Those who regard continuity of functional organisation as fundamental might argue that no break ever occurred. Others would maintain that a biological substrate may be essential to conscious experience. Neither position is established by the imagined experiment. The question is what would have to remain for each experience to continue belonging to the same subject.

The difficulty increases if we change the procedure. Suppose that, instead of progressively replacing the brain, we manage to reproduce its entire organisation in another substrate. The original remains alive, and a copy appears with the same memories, personality and abilities. Let us also suppose that both are conscious; reproducing that organisation does not, by itself, establish this. On that assumption, both would recognise the earlier biography as their own and would begin accumulating different experiences from that moment onwards. The functional resemblance might be perfect, but there would be two trajectories. Would we say that the original person had been duplicated, that a copy had inherited their history, or that personal identity allows some form of branching? We could compare their memories and behaviour, but the question of whether the original individual's subjective experience had continued in the new substrate would remain open.

Neuroprostheses let us study a specific aspect of this relationship between body and identity: asking someone about contact received through a device, or about a movement that begins to feel like their own. The thought experiments above extend the problem far beyond those findings.

The cockroach remains an organism whose locomotion is guided from outside. The electronics are not replacing its mind. It therefore offers a starting point for examining how functions are distributed, but no evidence that an identity can be transferred.

As we discussed in *Could We Copy a Human Mind?*, reproducing an ability and preserving the one who experiences it are different problems. *Eidos* explores what it would mean to bridge that gap for someone who wants to go on living. Today's interfaces let us approach one part of it through the experience of people whose bodies already incorporate technology.

One Final Reversal

For centuries, we used animals as a source of labour. Mechanisation replaced some of that effort with motors, while engineering learned to reproduce abilities such as flight, vision and locomotion. Biorobotics introduces another relationship: the organism itself can become a component of a machine. These ways of using animals and technology now coexist, each with its advantages and costs.

For many missions, a wholly artificial robot will remain preferable: units can be manufactured with more uniform characteristics, control may be more predictable, and there is no need to use a living being.

The cockroach in a diving suit makes the integration of organism and machine visible. Engineering retains the animal's body and adds what it needs to reach another environment. In a human neuroprosthesis, that integration serves the person using it: it may let them move a hand, feel contact or regain independence. The device enters a personal history that existed before it and may change through its use.

Someone who uses a prosthesis every day may eventually pay as little attention to its controller as we do to the neural circuits coordinating our fingers. Part of their activity will depend on a system built by others, with decisions and limitations of its own. That technical dependence will coexist with the independence the device allows them to regain.

> *Where do we remain, as more and more of what we perceive, decide and do comes to depend on the machine?*